CHAPTER 13

# STATISTICAL QUALITY CONTROL AND VITAL STATISTICS

**CHAPTER ESSENCE AT A GLANCE:**
Master SQC principles, 3-sigma control limits for variables (X-bar and R charts) and attributes (p and c charts), and demographic measures of mortality, fertility, and life tables.

## 4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Chance vs Assignable Causes of Variation:** Chance causes are random, inherent, and unavoidable, leaving the process in statistical control; Assignable causes are identifiable, systematic faults (tool wear, operator error) that must be detected and removed.

• **Control Charts for Variables (X-bar and R):** Used for measurable characteristics (diameter, weight); X-bar chart monitors process average; R (Range) chart monitors process dispersion; 3-sigma limits: UCL = Center Line + 3 * SE, LCL = Center Line - 3 * SE.

• **Control Charts for Attributes (p and c):** p-chart monitors fraction defective in samples of varying or fixed size; c-chart monitors count of defects per unit using Poisson model.

• **Demographic Vital Statistics Rates:** Measures population dynamics: Crude Birth Rate (CBR), Crude Death Rate (CDR), Infant Mortality Rate (IMR), Age-Specific Death Rate (ASDR), and complete Life Tables.

## ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\bar{X}\text{-Chart Limits} \Longleftrightarrow \text{CL} = \bar{\bar{X}}, \quad \text{UCL} = \bar{\bar{X}} + A_2\bar{R}, \quad \text{LCL} = \bar{\bar{X}} - A_2\bar{R}$$

*Three-Sigma Process Mean Control Boundaries*

$$R\text{-Chart Limits} \Longleftrightarrow \text{CL} = \bar{R}, \quad \text{UCL} = D_4\bar{R}, \quad \text{LCL} = D_3\bar{R}$$

*Three-Sigma Process Dispersion Range Boundaries*

$$c\text{-Chart Limits} \Longleftrightarrow \text{CL} = \bar{c}, \quad \text{UCL} = \bar{c} + 3\sqrt{\bar{c}}, \quad \text{LCL} = \max(0,\ \bar{c} - 3\sqrt{\bar{c}})$$

*Poisson Count of Defects Three-Sigma Boundaries*

$$\text{IMR} = \frac{\text{Infant Deaths under 1 Year}}{\text{Total Live Births}} \times 1000$$

*Infant Mortality Rate Key Public Health Index*

## CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Range and attribute charts, if calculated LCL is negative, it must be set to ZERO (LCL = 0) because range, fraction defective, and counts of defects cannot be negative.

• **Alert:** A process is in statistical control when all sample points fall within control limits and exhibit no non-random trend or run patterns.

• **Alert:** Crude Death Rate cannot be used to compare mortality across countries with different age structures; Standardized Death Rate (SDR) must be used instead.

## HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

**1. Distinguish between Chance Causes and Assignable Causes of variation in Statistical Quality Control. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **Chance (Random) Causes:** Inherent, minor, natural variations present in every production process despite best machine adjustments; individually small and economically unfeasible to eliminate. A process influenced only by chance causes is in statistical control. **[1.5 Marks]**

• **Assignable Causes:** Significant, systematic faults arising from identifiable sources like defective raw materials, machine tool wear, wrong operating temperature, or operator carelessness. Can be discovered and eliminated to bring the process back under control. **[1.5 Marks]**

**2. For 10 samples of size 5 each, the grand mean X double bar is 45 and the mean range $R_{bar}$ is 4.0. Construct the 3-σ control limits for X-bar chart and R chart (given for n=5: A2 = 0.577, D3 = 0, D4 = 2.115). [Score: 4 Marks]**

**Answer & Key Valuation Steps:**

• **X-bar Chart Control Limits:**<br/>• Center Line (CL) = X double bar = 45.<br/>• UCL = X double bar + A2 * $R_{bar}$ = 45 + 0.577 × 4.0 = 45 + 2.308 = 47.308.<br/>• LCL = X double bar - A2 * $R_{bar}$ = 45 - 0.577 × 4.0 = 45 - 2.308 = 42.692. **[2 Marks]**

• **R-Chart Control Limits:**<br/>• Center Line (CL) = $R_{bar}$ = 4.0.<br/>• UCL = D4 * $R_{bar}$ = 2.115 × 4.0 = 8.46.<br/>• LCL = D3 * $R_{bar}$ = 0 × 4.0 = 0. **[2 Marks]**

**3. In a manufacturing inspection, 25 carpets revealed a total of 100 defects. Calculate the 3-sigma control limits for a c-chart. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **Average Defects $c_{bar}$:** Total defects divided by number of units = 100 divided by 25 = 4.0 defects per carpet. **[1 Mark]**

• **Formula:** CL = $c_{bar}$ = 4.0; 3 * square root of $c_{bar}$ = 3 * square root of 4 = 3 × 2 = 6. **[1 Mark]**

• **Limits:** UCL = $c_{bar}$ + 3 * $\sqrt{(c_{bar})}$ = 4 + 6 = 10; LCL = $c_{bar}$ - 3 * $\sqrt{(c_{bar})}$ = 4 - 6 = -2. Since count of defects cannot be negative, set LCL = 0. Control limits: CL = 4, UCL = 10, LCL = 0. **[1 Mark]**

**4. Define: (a) Crude Death Rate (CDR), (b) Infant Mortality Rate (IMR). State their formulas. [Score: 3 Marks]**

**Answer & Key Valuation Steps:**

• **(a) Crude Death Rate (CDR):** Number of deaths occurring in a year per 1,000 estimated mid-year population. Formula: CDR = (Total Deaths divided by Total Mid-Year Population) * 1,000. **[1.5 Marks]**

• **(b) Infant Mortality Rate (IMR):** Number of deaths of infants under one year of age per 1,000 live births in a given year. Formula: IMR = (Infant Deaths under 1 year divided by Total Live Births) * 1,000. **[1.5 Marks]**

**5. What is a Life Table? State the meaning of the column expectation of life (e0). [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• **Life Table:** A tabular mathematical model exhibiting the complete mortality and longevity experience of a cohort from birth until death at each age. **[1 Mark]**

• **Expectation of Life (e0):** The average number of additional years a newborn infant is expected to survive under the prevailing age-specific mortality schedule. **[1 Mark]**

**6. State two criteria indicating that a manufacturing process is out of statistical control on a control chart. [Score: 2 Marks]**

**Answer & Key Valuation Steps:**

• **Criterion 1:** One or more plotted sample points fall outside the 3-sigma control limits (above UCL or below LCL). **[1 Mark]**

• **Criterion 2:** Non-random patterns appear within the control limits, such as a continuous run of 7 or more consecutive points on one side of the center line. **[1 Mark]**